

FEP Protocol Builder: Optimization of Free Energy Perturbation Protocols Using Active Learning

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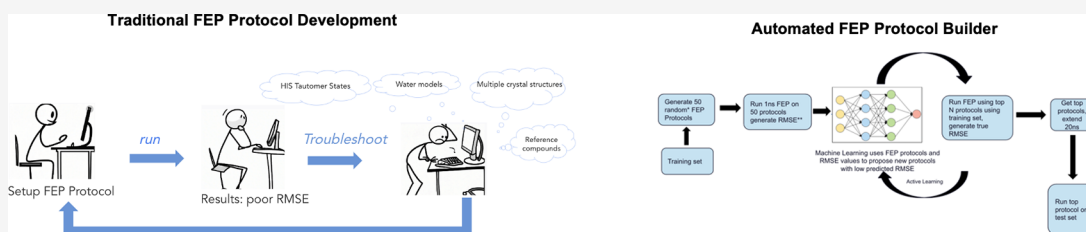
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ABSTRACT: Significant improvements have been made in the past decade to methods that rapidly and accurately predict binding affinity through free energy perturbation (FEP) calculations. This has been driven by recent advances in small-molecule force fields and sampling algorithms combined with the availability of low-cost parallel computing. Predictive accuracies of ~ 1 kcal mol⁻¹ have been regularly achieved, which are sufficient to drive potency optimization in modern drug discovery campaigns. Despite the robustness of these FEP approaches across multiple target classes, there are invariably target systems that do not display expected performance with default FEP settings. Traditionally, these systems required labor-intensive manual protocol development to arrive at parameter settings that produce a predictive FEP model. Due to the (a) relatively large parameter space to be explored, (b) significant compute requirements, and (c) limited understanding of how combinations of parameters can affect FEP performance, manual FEP protocol optimization can take weeks to months to complete, and often does not involve rigorous train-test set splits, resulting in potential overfitting. These manual FEP protocol development timelines do not coincide with tight drug discovery project timelines, essentially preventing the use of FEP calculations for these target systems. Here, we describe an automated workflow termed FEP Protocol Builder (FEP-PB) to rapidly generate accurate FEP protocols for systems that do not perform well with default settings. FEP-PB uses an active-learning workflow to iteratively search the protocol parameter space to develop accurate FEP protocols. To validate this approach, we applied it to pharmaceutically relevant systems where default FEP settings could not produce predictive models. We demonstrate that FEP-PB can rapidly generate accurate FEP protocols for the previously challenging MCL1 system with limited human intervention. We also apply FEP-PB in a real-world drug discovery setting to generate an accurate FEP protocol for the p97 system. FEP-PB is able to generate a more accurate protocol than the expert user, rapidly validating p97 as amenable to free energy calculations. Additionally, through the active-learning workflow, we are able to gain insight into which parameters are most important for a given system. These results suggest that FEP-PB is a robust tool that can aid in rapidly developing accurate FEP protocols and increasing the number of targets that are amenable to the technology.

INTRODUCTION

One of the primary goals in a drug discovery campaign is improving ligand binding affinity toward a given on-target, while simultaneously reducing binding affinity toward off-targets.^{1,2} Computational methods that can accurately predict binding affinity can significantly accelerate drug discovery projects by (a) reducing the number of compounds that have to be synthesized and assayed and (b) enabling the exploration of vast chemical space *in silico* to rapidly discover key molecules that advance the project.^{3,4} Free energy perturbation (FEP) calculations provide a rigorous, accurate approach to predicting binding free energies.^{5,6} FEP has recently become more accessible due to simultaneous advances in small-molecule force fields, molecular dynamics sampling methods, and GPU cloud computing.^{7–11}

A Free Energy Perturbation (FEP) protocol is defined as the input receptor and ligand structures alongside the settings of various simulation parameters involved in a free energy calculation (Table 1). To successfully deploy FEP technology on a drug discovery project, the primary requirement is to have a robust FEP protocol that can predict the experimental binding affinity with sufficient accuracy. In practice, we and others have found that acceptable accuracy for prospective

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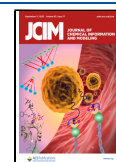


Table 1. Active Learning Optimization Parameters

parameters used in the active learning optimization	
FEP+	input values
receptor structure(s)	single or multiple protein receptors
ligand training/test set	label ligand membership in training or test set
custom core	SMARTS pattern(s)
production timescale (ns)	1
equilibration timescale (ns)	0.5, 1.0, 5.0
water models	SPC, TIP4P, TIP4PEW, TIP5P
water sampling	GCMC ON or OFF
lambda schemes	(12, 16, 24) or (16, 20, 28)
ligand REST	default, solvent, complex schemes
protein REST	randomly select 0, 1, 2, or 3 active residues
ligand enhanced sampling	torsion scaling ON or OFF
reference ligands	any two ligands in dataset
protein force field	OPLS3e or OPLS-AA-MM
ligand force field	FFBuilder default or advanced mode
HIS tautomer/protonation states	HID, HIE, HIP
GLN and ASN orientation	flipped orientation ON or OFF
assay top and bottom	upper and lower detection limits of experimental binding assay

application is a root-mean-square error (RMSE) less than 1.3 kcal/mol along with no evidence of convergence problems.¹² With recent improvements to FEP calculations, a large number of systems routinely produce accurate results with the default FEP protocol.^{7,13,14} Additionally, improvements in usability and automation^{7,15,16} have resulted in increases in the adoption of large-scale FEP calculations in drug discovery settings.^{13,17,18} For example, we have previously demonstrated how an accurate FEP protocol can rapidly accelerate a drug discovery project through the synthesis of fewer compounds by exploring billions of molecules *in silico*.^{19,20} Despite these compelling results, there are still a significant number of structurally enabled targets for which FEP is currently not amenable, oftentimes due to the lack of an accurate FEP protocol. In cases where the default FEP protocol cannot produce an accurate model,²¹ it requires the intervention of expert users to develop a working FEP protocol, which is often highly system-dependent and labor-intensive. There are numerous examples in the literature where significant human expert intervention was required to generate an accurate FEP protocol, and simple changes such as longer production timescales were insufficient to improve predictive performance.^{13,22} For instance, recent work found that certain water models work well for FEP calculations in polar sites of CYP450, while other water models work better for hydrophobic sites.²³ In terms of adequate sampling, Aldeghi et al. showed that equilibration times longer than 1 ns were necessary to produce accurate FEP predictions for the BRD4 system.²⁴ Enhanced sampling methods, such as replica exchange with solute tempering (REST),²⁵ can also be applied to further improve convergence and accuracy of FEP calculations. In multiple studies, Wang et al. showed that the inclusion of select binding site residues in the REST region improves the accuracy of FEP predictions for the thrombin and CDK2 system.^{8,26} Cole et al. reported similar findings for HIV reverse transcriptase (HIVRT).²⁷ Fratev et al. found that FEP performance was improved by including the entire ligand in the REST region for several systems.²⁸ Concerning receptor

structures, Schindler et al. reported that the selection of loop conformation for the kinesin Eg5 protein had a significant effect on the FEP accuracy.¹³ Similarly, Steinbrecher et al. findings strongly suggested that the selection of protein receptor structures can significantly affect FEP model performance for the JAK2, HSP90, and p38alpha MAP kinase systems.²⁹ Matricon et al. reported that manual selection of protonation states of multiple histidine residues, alternative conformations of His264 and Glu169, and the TIP3P water model resulted in good retrospective FEP performance for the A2A adenosine receptor binding site.³⁰ Accurate description of water molecules interactions and dynamics around receptor binding sites are also of key importance for the reliable prediction of protein–ligand binding affinities. As shown by Bos et al., the use of Grand Canonical Monte Carlo sampling (GCMC)^{31,32} to improve water sampling for the solvent occluded binding site of D-Amino-Oxidase was important to produce high-quality FEP prospective predictions.¹⁹ Arriving at these aforementioned optimized FEP protocols usually involves several iterations of manually generating FEP protocols, running the FEP calculations, analyzing the results, and refining subsequent FEP protocols. This process is generally labor- and time-intensive and usually only explores a small portion of the parameter space, oftentimes guided by intuition versus a more systematic data-driven approach. This relatively long timeframe is often not in line with tight drug discovery timelines, where design–make–test–analyze (DMTA) cycles are usually on the order of 2 weeks.¹³ In a recent large-scale validation of FEP in real-world drug discovery settings, Schindler et al. acknowledge that they expect prospective accuracy to improve if the FEP protocols were further optimized, but manual optimization would take far longer than project timelines.¹³ To address these limitations, we developed FEP Protocol Builder (FEP-PB), which is able to significantly accelerate FEP protocol generation via (1) fully automated generation, testing, and validation of FEP protocols using active learning;^{33,34} (2) incorporation of training/test set splits to avoid overfitting; and (3) employing a data-driven workflow to rapidly converge on accurate FEP protocols that not only have good RMSE values but also good convergence properties. As a real-world test case, we use the Myeloid cell leukemia sequence 1 (MCL1) system, where default FEP settings were unable to produce a protocol with suitable accuracy. FEP-PB is able to rapidly generate accurate protocols with RMSE <1.3 kcal/mol with good convergence properties. We also apply FEP-PB to real-world drug discovery settings in the validation of the protein p97 system where default settings were unable to generate an accurate protocol. For this system, we also compare the performance of FEP-PB against the expert user. We find that FEP-PB is able to generate a more accurate FEP protocol than the expert user and rapidly validate p97 as a target that is amenable to free energy calculations

METHODS

System Preparation. The starting receptor structures for FEP-PB were the PDB structure with ID 4HW3³⁵ for MCL1 and a proprietary Cryo-EM structure for p97, similar to the publicly available Cryo-EM PDB structure with ID 7RLI. The p97 structure contained an allosteric ligand bound at a site distal from where orthosteric ligands bind. It was not clear if the presence of this allosteric ligand would improve FEP performance so two receptor structures were prepared, one

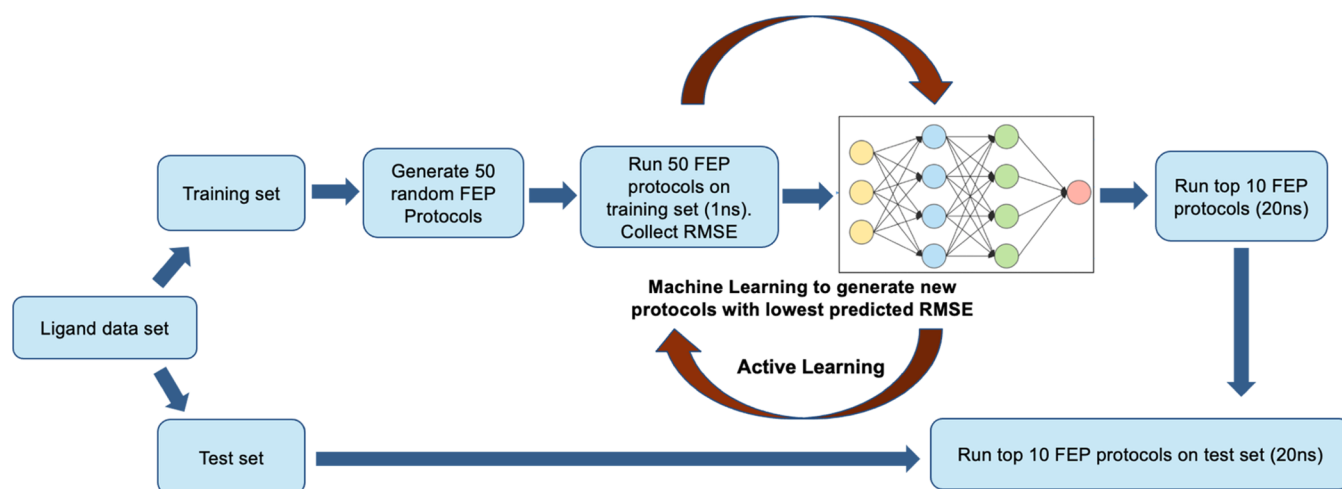


Figure 1. FEP-PB active learning workflow.

with the allosteric ligand present and one with the allosteric ligand absent. The OPLS3e force field was used for the proteins and the ligands.³⁶ The proteins were prepared using the Protein Preparation Wizard during which the initial protonation states were assigned assuming a pH of 7.0.³⁷ The systems were solvated in a water box with a buffer width of 5 Å for the complex simulations and 10 Å for the solvent simulations. The systems were relaxed and equilibrated using the default Desmond relaxation protocol (Supporting Information). The entire system with the solute molecules restrained to their initial positions was first minimized using the Brownie integrator and then simulated at 10 K using an NVT ensemble followed by an NPT ensemble. The system was then simulated at room temperature using the NPT ensemble with the restraints retained. The Bennett acceptance ratio method (BAR) was used to calculate the free energy.³⁸ Errors were estimated for each free energy calculation using both bootstrapping and the BAR analytical error prediction.

Ligand Datasets. For the MCL1 system, 42 ligands were selected from Friberg et al.³⁵ (Figure S1). For the p97 system, 15 ligands were selected from Zhou et al.³⁹ (Figure S3). Experimental binding affinity data are expressed as Gibbs Free Energy (ΔG) of binding and were obtained from the published IC_{50} or K_i (inhibition constant) values. Ligand structures were processed using LigPrep to enumerate all stereoisomers, and protonation states were assigned using Epik. OPLS3e force field torsion parameters for each ligand were generated using the Force-Field Builder module (FFBuilder).^{36,40}

FEP-PB Active Learning Workflow. Figure 1 provides an overview of the automated FEP Protocol builder workflow. For the first step, the ligand dataset is randomly split into equally sized training and test sets. The train-test split procedure is appropriate to increase the probability of producing robust models and also to avoid potential overfitting and data dredging.^{41,42} The training set is used to run the active-learning workflow, while the test set is used to evaluate the performance of the top optimized protocols. Using the input structures and information about the parameter space to explore, the workflow generates 50 (or a user-defined number) random FEP protocols. The FEP protocols are generated and stored using the javascript object notation (JSON) file format.⁴³ The protocols are randomly generated to help ensure the parameter combinations that are sampled are representative of the entire distribution of possible protocols. Each random protocol is

defined by assigning a unique value for each one of the parameters displayed in Table 1. Once the random protocols are generated, an FEP simulation calculation is run using each random protocol. The FEP simulation production timescale for the active learning portion of the workflow is set to 1 ns. This is done primarily to (1) keep compute costs low, and (2) have the workflow complete in a reasonable timeframe in line with real-world drug discovery projects. After each FEP run is completed, edgewise RMSE values comparing predicted free energies of binding with their respective experimental values are collected. The RMSE values along with their respective protocols are then used to train a machine learning (ML) model. The ML model is built with TPOT,⁴⁴ a Python automated machine learning tool (AutoML) tool that uses genetic programming to optimize machine learning pipelines. The workflow then executes a random search through the FEP protocol parameter space followed by the scoring of generated FEP protocols using the ML model. The top 50 protocols with the lowest ML predicted RMSE values are selected. Next, the workflow automatically sets up and runs FEP calculations based on the top 50 ML-selected FEP protocols. In this work, we repeated these steps two times. After the final iteration, the 10 top FEP protocols are selected based on the lowest RMSE computed from the 1 ns FEP calculations. To ensure the lowest RMSE scoring protocols also have good convergence properties, we extended the simulation time of the top 10 protocols to 20 ns. Finally, the top extended FEP protocols with the lowest RMSE and good convergence properties are run against the test set. The FEP protocol with the best performance on the training set and acceptable performance against the test set (RMSE <1.3 kcal/mol) is selected as the overall top-performing protocol. Although RMSE is the metric used here-in to select among the optimized models, we would like to also highlight the protocol optimization machinery introduced here can utilize, in principle any objective function to guide the optimization including MUE, R^2 , Kendall–Tau rank correlation coefficient, classification accuracy as well as complex composite measures combining these various measures of calculation quality.

FEP+ Parameters for Active Learning Optimization. During protocol optimization, FEP-PB is designed to take into account both FEP simulation parameters and receptor structural modifications, such as multiple receptor structures, residue orientations, and protonation states. The 1 ns

protocols are used to efficiently search the parameter space to select for models that have a higher probability of good performance at the longer timescale (20 ns). Table 1 outlines the FEP parameters explored during the active-learning workflow. Below is a short description of each parameter and how it relates to the overall FEP protocol accuracy.

Receptor Structure. The choice of the initial ligand–protein structure can be of key importance to the FEP protocol accuracy, especially for more dynamic systems.⁴⁵ Given an ensemble of structures (which can be derived experimentally or from *in silico* approaches), it is not often straightforward to know which structure will give the optimal FEP performance, especially for systems that display a high degree of conformational flexibility. Therefore, FEP-PB accommodates one or more protein–ligand complex structure files containing a specific conformation of the protein target and a set of congeneric ligands bound to the receptor binding site. The ligand binding poses can be varied for each specific receptor structure. This allows for active learning to select the optimal receptor structure and ligand binding modes that will produce the most accurate FEP protocol.

Custom Core. In most cases, FEP calculations involve the perturbation of a peripheral functional group or the central part (core) of the ligand. In the FEP+ implementation, perturbation pathways are automatically generated using a mapping algorithm as described by Wang et al.⁷ Compounds with high FEP similarity scores are connected by edges, which represent alchemical transformations between connected ligands. FEP similarity score for a pair of ligands uses a combination of 2D and 3D features such as their maximum common substructure (MCS) and the quality of their three-dimensional alignment in the receptor binding pocket.⁷ The combination of these features determines which atoms of the ligands are mutated during the alchemical transformation. The *custom core* feature in FEP+ enables control of which atoms are included in the mutation region of the molecule by specifying the core of the ligand. The *core*, in this context, corresponds to the ligand atoms that are not perturbed along the FEP calculation. Multiple custom cores can be supplied in the SMARTS format,⁴⁶ alongside using the default assignment.

Equilibration Timescales. The equilibration phase is generally used to bring the system to the desired temperature and pressure conditions. It is difficult to ascertain *a priori* what the appropriate equilibration time is for a given system, often resulting in either equilibration times that are unnecessarily long, wasting compute resources, or too short, which results in simulations that do not accurately represent the system of interest.⁴⁷ Therefore, we allow the active-learning workflow to select from a variety of timescales for the equilibration phase of the free energy calculations.

Water Models. Due to the impact of water thermodynamics on small-molecule binding,⁴⁸ the application of different water models in FEP calculations can significantly impact the accuracy of the results. The workflow allows for the inclusion of several popular water models.^{49–52}

Water Sampling. Accurate modeling of water molecules plays a key role in the accuracy of free energy calculations. Often the most important water molecules are those that occupy the binding site, particularly those which are kinetically trapped and cannot exchange with bulk solvent within the relatively short molecular simulation timescales.^{53–55} Grand Canonical Monte Carlo (GCMC) sampling allows for more

accurate water sampling in buried pockets.^{31,32,56} We allow the active-learning workflow to turn this approach on or off.

Lambda Schemes. During FEP calculations, the relative free energy between a pair of ligands is estimated by performing a series of molecular dynamics simulations at different coupling parameter (lambda) values.⁵⁷ This coupling parameter is a crucial part of the FEP approach as it allows the gradual transformation of one molecule into another. Moreover, the lambda parameter plays a critical role in FEP convergence and accuracy. In this work, a lambda scheme refers to a set of values that are applied to specific types of alchemical transformation. For instance, the “12, 16, 24” lambda scheme refers to r-group, core hopping, and charge perturbations, respectively.

Ligand and Protein REST. The replica exchange with solute tempering (REST) method carries out replica exchange simulations in which the potential energies of a selected subsystem have been scaled in a way that mimics the application of a locally higher temperature as one ascends the replica ladder but leaves the rest of the system at the desired temperature. In this way, the number of replicas required depends only on a small subset of the total system degrees of freedom. It has been shown that the selection of protein residues to place in the REST region can significantly improve or degrade performance.^{25–27,58} Since this is highly system-dependent, we allow the active-learning workflow to select up to three binding site residues to place in the REST region. The binding site is defined as any residues within 5 Å of a representative ligand. For the ligand REST, this approach facilitates sampling of ligand binding modes that are separated by high free energy barriers and ensures that computed free energy changes are considerably less dependent on the starting conditions and the chosen mutation pathway. We allow the active-learning workflow to select from three ligand REST settings (1) Default: For a perturbation between a pair of ligands, the region of the ligand directly involved in the mutation is included in the REST region. If the number of heavy atoms in the perturbed functional group is less than the cutoff value (25 heavy atoms), then the algorithm tries to include one more rotatable bond until the total number of heavy atoms exceeds the cutoff value;⁷ (2) Complex: default, Solvent: all. The entire ligand is placed in the REST region during the solvent leg of the simulation while the default setting is applied in the complex leg, and (3) Complex: all, Solvent: all. The entire ligand is placed in the REST region during the solvent and complex leg of the simulation.

Ligand Enhanced Sampling. In typical molecular dynamics simulations, the ligand conformational space is sampled by rotating internal degrees of freedom. Unfortunately, the high energy barriers of certain torsion angles can cause a limited sampling of certain conformations, which can result in subsequent inaccuracies in the free energy calculations. Ligand torsion scaling addresses this problem by selectively scaling energy barriers on the aforementioned torsion angles, allowing better sampling of conformational space. The use of torsion scaling is highly dependent on factors such as the conformational flexibility of the ligand as well as the flexibility of the protein binding site. We allow the active-learning workflow to turn this option on or off.

Reference Ligands. Reference ligands in free energy calculations are ligands that are used as reference points to compute the free energy difference between two ligands or a ligand–protein complex. Reference ligands are useful in

accounting for systematic errors as well as normalizing the predicted results. From large sets of congeneric ligands, it is often unclear which molecules to select as reference ligands, and this can have significant effects on the accuracy of the predictions.¹⁶ Therefore, we allow the active-learning workflow to select the optimal pair of reference ligands from a larger set of ligands provided by the user.

Ligand/Protein Force Field. Accurate free energy predictions are highly dependent on the accuracy of the potential energy function. The potential energy of the system of interest is calculated using a molecular mechanics force field, which is an approximation of the quantum mechanical forces. The level and implementation of these approximations can vary between different force field implementations, and it is not always clear which implementation is the most accurate *a priori*.^{36,59} We allow active learning to select between two implementations of the protein force field, OPLS3e, and OPLS-AA-M. The OPLS-AA-M incorporates new dihedral parameters into the OPLS-AA force field.⁵⁹ We also allow the FEP protocol builder workflow to evaluate different versions of the small-molecule force fields, the default and advanced versions of the small-molecule force field. The default small-molecule force field consists of the parameters included in the official release of OPLS3e.³⁶ Since the coverage of OPLS3e is not complete, we observed that for some ligands used in this work, there was at least one missing torsion parameter. For the default mode, we deploy FFbuilder⁴⁰ to provide force field parameters only for torsions that were not represented explicitly or by atom-type equivalency in the OPLS3e force field. Alternatively, for the advanced mode, we deploy FFbuilder to recalculate force-field parameters for all torsions of the molecule, regardless if the torsion was represented or not in the OPLS3e force field.

HIS Tautomer and Protonation States. The imidazole side chain of histidine can exist in different tautomeric forms which have different chemical and physical properties. The correct assignment of these forms can significantly impact the accuracy of free energy calculations. In general, x-ray crystallography cannot distinguish between different tautomeric forms of histidine within a protein structure. Therefore, we allow active learning to sample structures with the histidine proton at the epsilon position (HIE), the delta position (HID), and the double-protonated tautomer (HIP) where both nitrogen atoms of the imidazole ring are protonated.

GLN and ASN Orientation. The side-chain amide oxygen and nitrogen atoms of asparagine and glutamine residues are indistinguishable at standard crystallographic resolution, and can routinely be assigned incorrectly.¹⁶ Incorrect orientations of asparagine or glutamine residues have little effect on the measured quality of a crystal structure, but can significantly destabilize a protein during a molecular dynamics simulation. Additionally, the energetic barrier to correctly rotate these groups might be too high to achieve during the simulation. Therefore, we sample different starting orientations of these groups in the FEP protocol generation.

Assay Top and Bottom. Oftentimes, significant amounts of experimental binding data are reported at the highest (top) or lowest (bottom) concentration of the ligand in the experimental assay. This data is traditionally excluded from the validation of free energy calculations since the exact experimental value is outside the detection limit of the experimental assay. Here we use a correction function such that if the FEP predicted value is greater than the bottom of the assay or lower than the top of the assay, it is considered a

perfect prediction (eq 1). This allows for the active-learning workflow to build an accurate model based on both quantitative and qualitative data. This is particularly useful for generating models that accurately predict compounds that are very weak binders or inactive since one of the main purposes of FEP calculations is to enrich for more potent compounds. This parameter was not applied to the systems in this study since all of the available experimental data for both systems was quantitative. The correction function for experimental data at the top or bottom of the assay detection limit is shown as eq 1.

$$\text{RMSE} = \sqrt{\sum_{i=1}^n \frac{\text{qualified error}(x_i, \hat{x}_i)}{n}}$$

$$\text{qualified error} = \begin{cases} 0 & \text{if } x \leq \hat{x} \text{ and } \hat{x} = \text{assay}_{\text{low}} \\ 0 & \text{if } x \geq \hat{x} \text{ and } \hat{x} = \text{assay}_{\text{high}} \\ (x_i - \hat{x}_i)^2 & \text{otherwise} \end{cases} \quad (1)$$

Feature Importance of the FEP Protocols. A benefit of running the active-learning workflow over the FEP protocol space is that the individual protocols can then be run through a feature importance technique. This can give a deeper understanding of the relative importance of different parameters toward the overall accuracy of the FEP protocol. A popular technique for determining local and global feature contributions are Shapley values.⁶⁰ In this framework, one calculates the average marginal contribution of a feature across all possible feature coalitions. These marginal contributions on a sample-by-sample basis sum up to the difference between a sample's model prediction and the average prediction over the training set. While these Shapley feature contributions are on a sample basis, one can then average them across all samples to understand a feature's importance over a whole dataset.

RESULTS AND DISCUSSION

MCL1. MCL1 is involved in the regulation of apoptosis and is considered a pro-survival protein. Orthosteric inhibitors of this target are a viable therapeutic strategy for various cancers (Figure 2).⁶¹ This target has historically proven challenging for developing an accurate free energy model.⁷

Traditionally, the aforementioned poor FEP performance would be followed up with labor-intensive manual debugging to generate a more predictive FEP protocol. Here we instead apply FEP-PB to more rapidly and rigorously generate a predictive model. The settings used for FEP-PB were 50 protocols per active learning cycle, with three iterations of the active-learning workflow. Figure 3 displays the RMSE values associated with each of the protocols generated by FEP-PB during the active-learning workflow. Protocols 1–50 have a wide range of RMSE values, with some protocols returning RMSE values below 1.3 kcal/mol (colored in blue). A number of FEP protocols also showed poorer predictive power with RMSE above 1.3 kcal/mol (colored in red). This is expected since in the first iteration of the active learning workflow, the workflow generates a random set of 50 FEP protocols by randomly selecting parameter values from the defined parameter space (Table 1). Protocols 51–100 have a greater number of lower RMSE protocols compared to the first iteration cycle. This improvement is due to the first round of

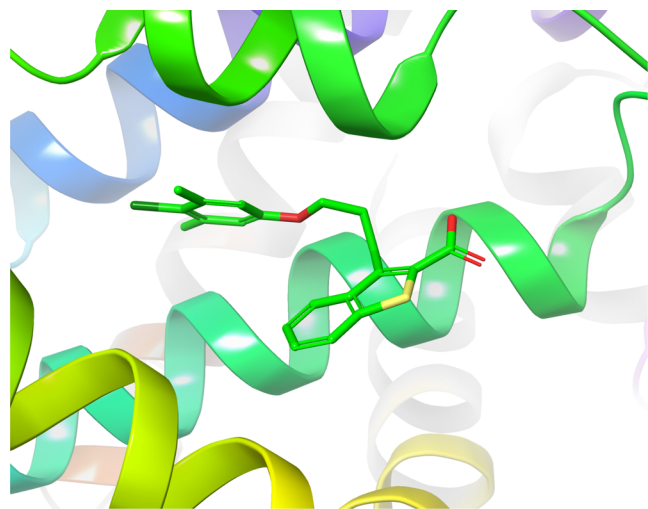


Figure 2. Inhibitor bound to the active site of MCL1 (PDB ID: 4HW3).

the active-learning workflow, which uses the output from the prior iteration to build an ML model for the subsequent iteration. The improved ML model is then able to identify the most relevant features from the FEP parameter space, and as a consequence, generate FEP protocols with higher predictive power. After another round of the active-learning workflow, the workflow successfully generated many more protocols with RMSE values lower than 1.3 kcal/mol. This suggests that the active-learning workflow is able to successfully discern which combination of parameters results in the FEP protocols with the lowest RMSE values.

Figure 4a illustrates the performance of the default FEP protocol on the training set (RMSE 1.62 kcal/mol), which may be insufficient for prospective application. Figure 4b displays the performance of the best FEP-PB model on the training set from the three iterations of the active-learning workflow (RMSE 1.10 kcal/mol), which is significantly better than the default protocol (Table S1). Extending this protocol out to 20 ns maintains good performance (Figure 4c). Moreover, the FEP-PB protocol displays a similar performance on the test set (Figure 4d). This increases confidence in the robustness of the

FEP-PB protocol since these ligands were not seen by the active-learning workflow.

Table 2 outlines the performance of the top FEP-PB protocol on the test set at 1 and 20 ns simulation time. We observed a slight increase in RMSE upon extending the protocols from 1 ns to 20 ns, which is expected since many of the protocols at 1 ns are not fully converged. It is possible that the RMSE could also remain the same or decrease upon extending the timescale from 1 to 20 ns. This is exemplified in Table 3, where we observed several convergence warnings for the top protocol at 1 ns, primarily due to the shorter production timescale (see the Supporting Information for a detailed explanation of convergence warnings). Upon extension of the top protocol to 20 ns, we see that the majority of these convergence warnings are resolved, suggesting that this model is suitable for prospective use. Similar trends are seen for all of the top-performing models for this system (Tables S1 and S2).

Overall performance of the top FEP-PB model for all ligands (i.e., training and test set combined) further demonstrates that the best-performing protocol is suitable for prospective application (Figure 5b) and significantly better than the default protocol (Figure 5a). We also observed a significant decrease in the number of large outliers (RMSE > 2 kcal/mol) in the FEP-PB top protocol relative to the default protocol (Figure 5).

Table 4 gives an overview of the parameters of the top FEP-PB protocol. Only parameters that vary from default values are shown. Interestingly, only a relatively small number of parameters have to be varied relative to default settings in order to obtain good performance. Nevertheless, due to the complexity and interplay between molecular structures, simulations, and energetics, it is often not straightforward to intuit which parameters are important for a system of interest. FEP-PB offers a rapid and rigorous approach to identifying these key parameters and generating accurate FEP protocols.

Feature Importance for the MCL1 System. An additional benefit of running the FEP-PB workflow is scientists can look at trends in the data and gain insight into key FEP parameters of importance in their system of interest. In Figure 6, we show the distributions of the nine largest average Shapley values over the MCL1 dataset when calculated by training an XGBoost⁶² model over all of the FEP protocols which were

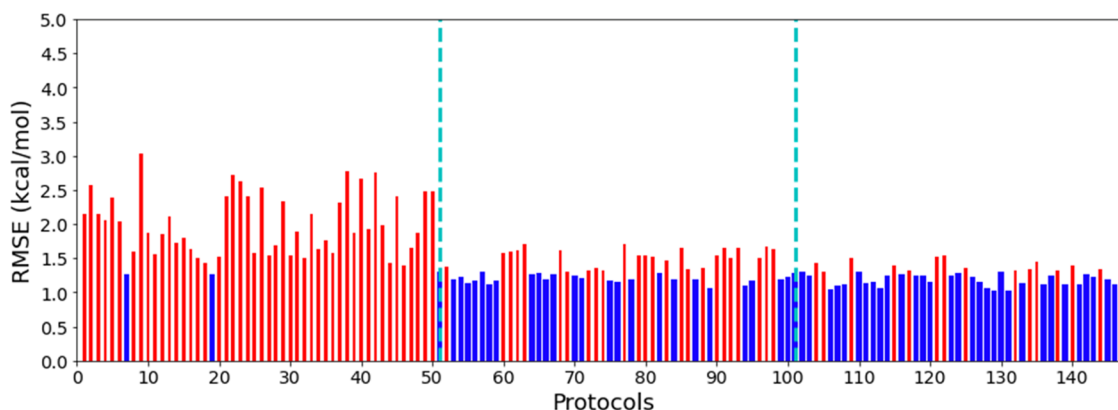


Figure 3. FEP Protocol Builder results for the MCL1 system after 3 iterations of the active-learning workflow. RMSE values were calculated from predicted FEP and experimental free energy of binding. Blue bars represent protocols with RMSE values below 1.3 kcal/mol, while red bars represent protocols with RMSE values above 1.3 kcal/mol. Vertical cyan dashed lines mark each active learning iteration cycle, which is characterized by 50 FEP protocols.

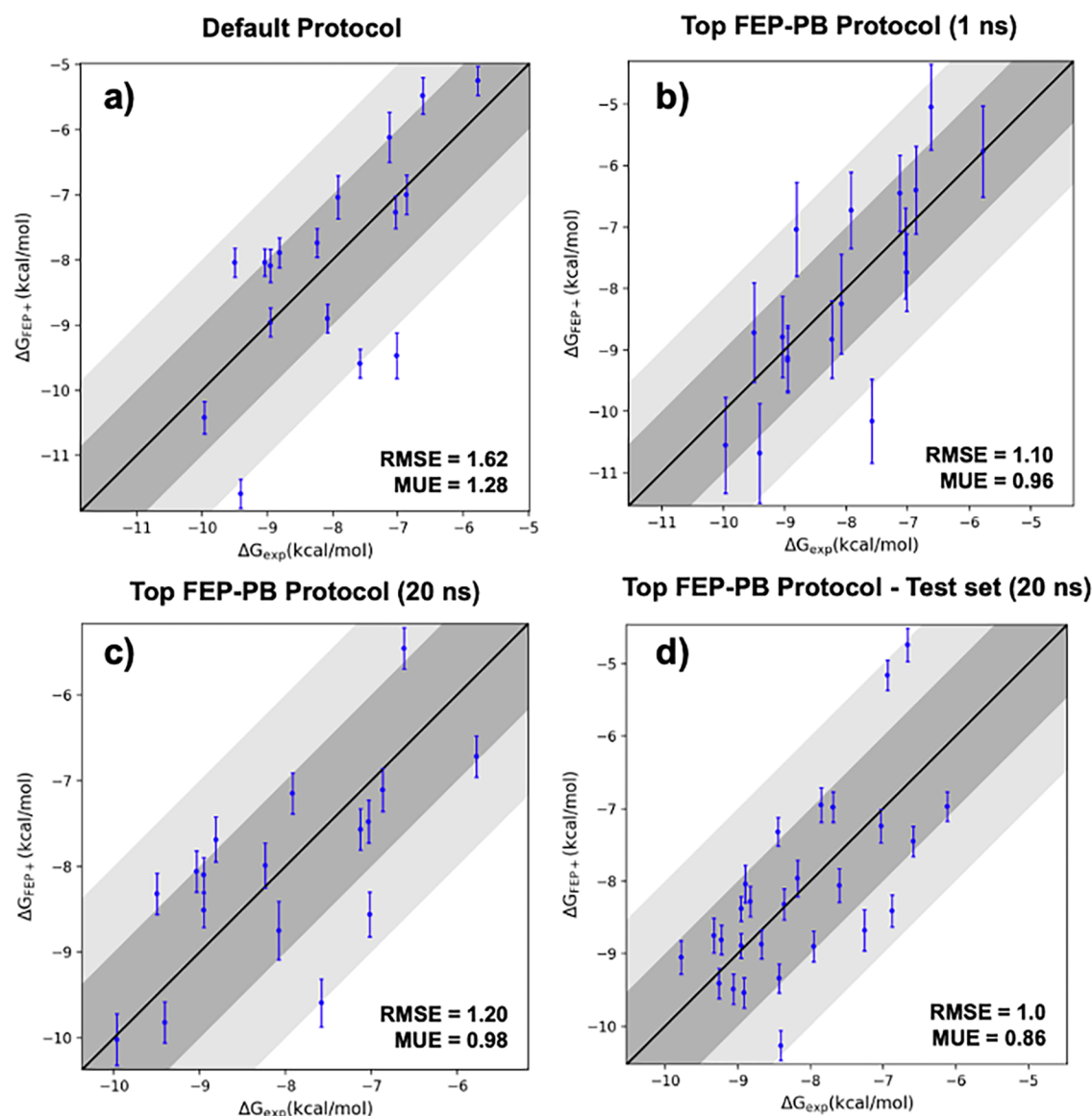


Figure 4. FEP performance of (a) default FEP Protocol on the training set, (b) top FEP-PB protocol on the training set (1 ns simulation time), (c) top FEP-PB protocol on the training set (20 ns simulation time), and (d) top FEP-PB protocol on the test set (20 ns simulation time). Predictions within 1 and 2 kcal/mol of the experimental affinity are highlighted by dark and light gray areas, respectively.

Table 2. FEP-PB Top Protocol Statistical Performance on the Test Set at 1 and 20 ns Simulation Times

production timescale (ns)	R^2	RMSE (edgewise)	MUE (edgewise)	RMSE (pairwise)	MUE (pairwise)
1	0.49	1.10	0.96	1.52	1.24
20	0.44	1.20	0.98	1.37	1.12

run. The most important features are: sampling Val 253 in the REST region, not using reference ligand 64 and instead using reference ligands 26, 34, or 37, using the TIP5P water model, and not using GCMC water sampling.

p97. The p97 protein, also known as valosin-containing protein (VCP) plays a critical role in protein quality control, including degradation and turnover, as well as the removal of misfolded and damaged proteins from the endoplasmic reticulum (ER).⁶³ Inhibitors of p97 have therapeutic applications in cancer since p97 plays a key role in the degradation of oncoproteins and regulation of cell cycle progression (Figure 7).^{64,65} Computationally validating this target first required developing an accurate FEP protocol. The default protocol was applied to the available literature SAR dataset (Figure S3), which resulted in relatively poor

Table 3. Convergence Properties of the FEP-PB Top Protocol on the Test Set at 1 and 20 ns Simulation Times

production timescale (ns)	number of edges with convergence warnings				
	FEP_ERROR	energy convergence	REST exchange	cycle closure convergence	cycles with high hysteresis
1	20	31	0	13	50
20	0	0	0	3	3

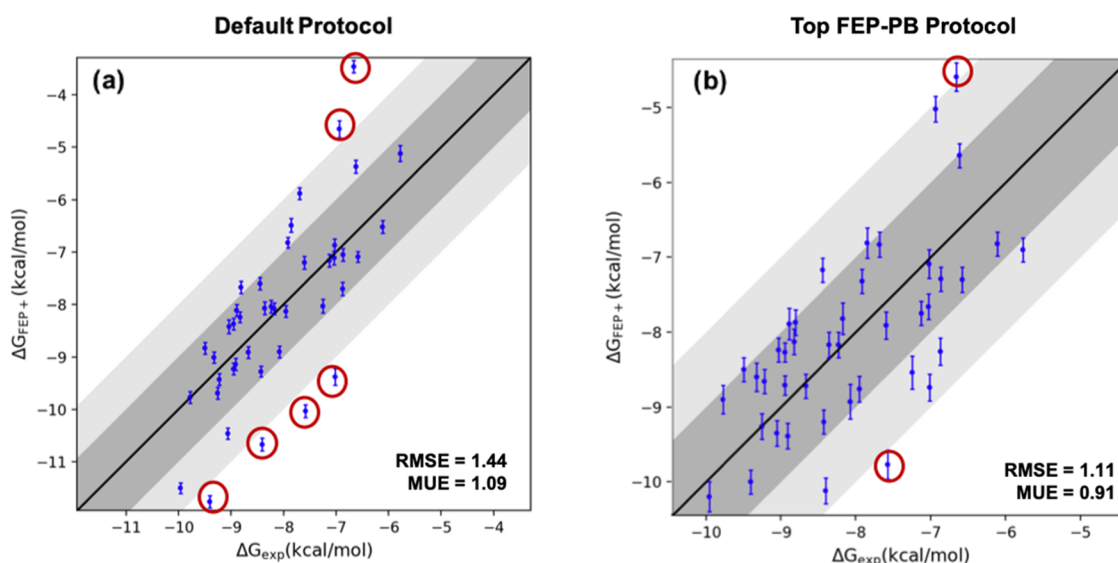


Figure 5. FEP performance for all ligands using (a) the default FEP protocol and (b) the top FEP-PB protocol. Large outliers (RMSE > 2 kcal/mol) are circled in red. Predictions within 1 and 2 kcal/mol of the experimental affinity are highlighted by dark and light gray areas, respectively.

Table 4. Parameters for the Top FEP-PB Protocol for the MCL1 System^a

MCL1 top FEP-MB protocol parameters	
FEP+	input values
water models	TIP5P
lambda schemes	(16, 20, 28)
ligand REST	entire ligand in REST solvent
protein REST	Val 253, Leu 235
HIS tautomer/protonation states	HID 224, HID 252
reference ligands	ligand 37, ligand 45

^aAll parameters not shown are set to default values.

performance for prospective utilization (RMSE 1.33 kcal/mol) (Figure 8a). This is a relatively challenging SAR set for FEP since there are simultaneous perturbations to several regions of the molecules (Figure S3). Consequently, we enlisted an FEP expert user to manually generate a more accurate FEP

protocol. At the same time, we also deployed FEP-PB to compare both approaches. Only 1 week was available for both approaches to attempt to generate a predictive FEP protocol. FEP-PB was run in a similar fashion to the MCL1 system and detailed results of the active-learning workflow can be found in the Supporting Information (Figures S4 and S5). Figure 8 outlines the performance of the (a) default protocol, (b) the expert model, and (c) the top FEP-PB model on the entire p97 dataset. The expert was able to develop a protocol that slightly improved FEP performance (RMSE 1.25 kcal/mol) but still contained several significant outliers with RMSE > 1.3 kcal/mol. FEP-PB produced the most accurate protocol (RMSE 1.08 kcal/mol) and is suitable for prospective use. Interestingly, the receptor structure without the allosteric ligand gives the best FEP performance and is consistently selected by the active-learning workflow in the top-performing FEP-PB protocols (Figure S4). Comparing the expert and FEP-PB protocols, the expert user was able to discern the importance of

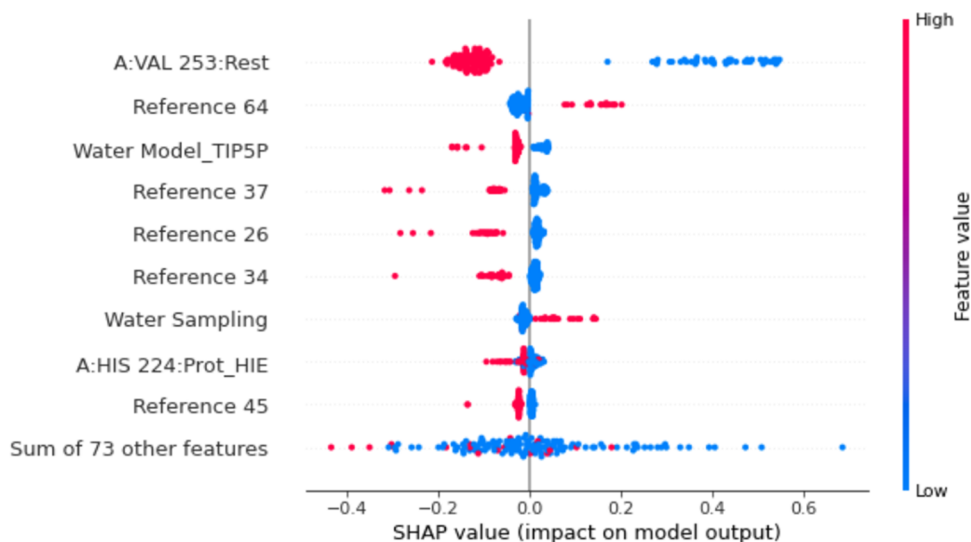


Figure 6. Distributions of the nine largest average absolute Shapley values over the MCL1 dataset.

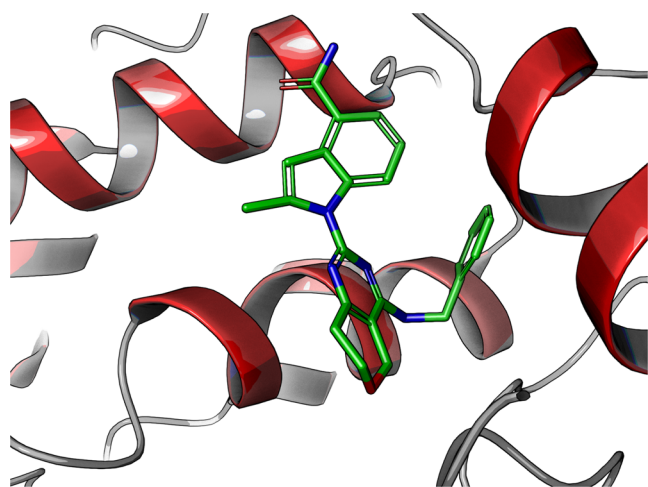


Figure 7. Inhibitor bound to the active site of p97 (PDB ID: 7RLI). Perturbations occur simultaneously at the indole and phenyl r-groups along with changes to the tetrahydroquinoxaline core.

extending the lambda scheme and placing the entire ligand in the REST region but selected the suboptimal receptor structure and also did not modify the equilibration time, protein REST, and ligand force field parameters (Table 5).

Table 5. Parameters for the Expert User Protocol for the p97 System^a

p97 expert user protocol parameters	
FEP+	input values
receptor	receptor containing allosteric ligand
lambda schemes	(20, 20, 24)
ligand REST	entire ligand in REST solvent
reference ligands	ligand 35, optimal map

^aAll parameters not shown are set to default values.

Modifications to the aforementioned parameters resulted in the most accurate protocol from FEP-PB (Table 6). It is important to emphasize that these results do not reflect a lack of expertise of the human user, but instead, illustrate the intractability of manually exploring the FEP parameter space in

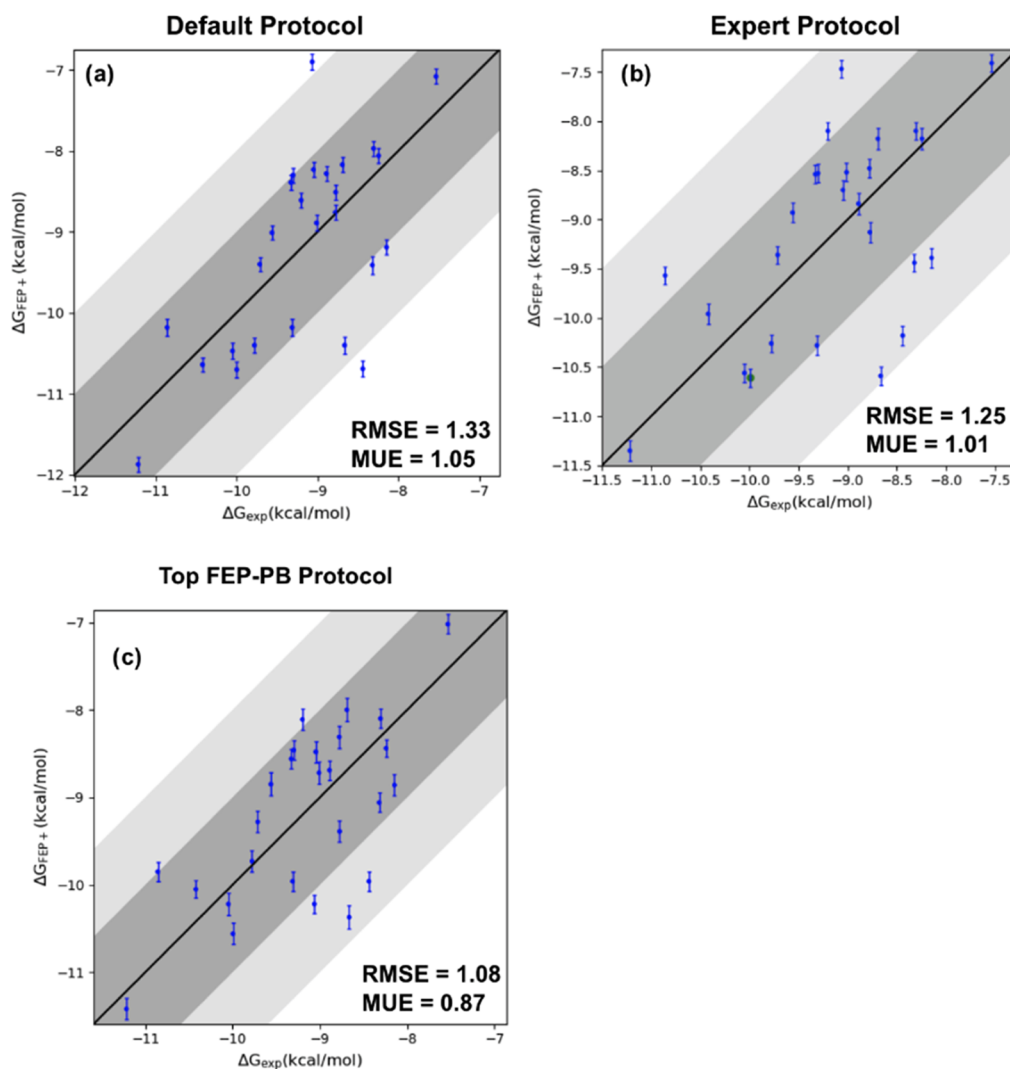


Figure 8. FEP performance of (a) default FEP protocol on the entire dataset, (b) expert protocol on the entire dataset, and (c) top FEP-PB protocol on the entire dataset. Predictions within 1 and 2 kcal/mol of the experimental affinity are highlighted by dark and light gray areas, respectively.

Table 6. Parameters for the Top FEP-PB Protocol for the p97 System^a

p97 top FEP-PB protocol parameters	
FEP+	input values
receptor	receptor without allosteric ligand
water model	TIP4PEW
lambda schemes	(16, 20, 28)
equilibration time	0.5 ns
ligand REST	entire ligand in REST solvent
protein REST	Arg 668, Trp 475
ligand force field	force-field builder advanced mode ON
reference ligands	ligand 28, ligand 72

^aAll parameters not shown are set to default values.

a rigorous manner within standard drug discovery timelines. This is in agreement with the work of Schindler et al. who found that the relatively tight drug discovery timelines prevented them from carrying out extensive optimization of FEP protocols for systems where the default FEP protocol performed poorly.¹³

CONCLUSIONS

In the past few years, we have observed significant and simultaneous improvements in compute power, structural biology, and the predictive accuracy of free energy perturbation calculations.^{66,67} To fully take advantage of these improvements, we developed FEP-PB, which employs the active-learning workflow to rapidly develop accurate FEP protocols for biological targets of interest. We demonstrate the utility of FEP-PB by applying it to the optimization of FEP protocols for a pharmaceutically relevant target, MCL1, where the default settings do not produce suitable model accuracy. FEP-PB is able to generate protocols with good accuracy and convergence that can be applied prospectively. When applied head-to-head against the expert user, FEP-PB is able to generate a more accurate protocol for the p97 system, requiring only ~1 week to complete all calculations using cloud computing resources, which is in line with common drug discovery timelines. Importantly, this workflow is designed to be carried out with limited human intervention, which opens up the possibility of validating many more systems in a much shorter timeline. For example, we can now accelerate the ability to rapidly validate off-target models for modeling ligand selectivity with FEP.⁶⁸ We have used FEP-PB extensively in our internal drug discovery projects and found it able to generate low RMSE protocols for several difficult targets. These FEP protocols also perform well when applied prospectively for these targets. With the advent of AlphaFold,⁶⁹ we have also regularly been incorporating these protein receptor models into the FEP-PB active learning workflow, further widening the pool of targets amenable to FEP. Furthermore, we are collecting FEP-PB active-learning workflow data from all of these projects to better decipher which settings consistently improve FEP model performance across many targets. By calculating the feature importance across these diverse targets, the goal is to eventually arrive at a more robust default protocol. Overall, we expect that FEP-PB will greatly increase the number of targets that are amenable to free energy perturbation calculations, allowing rapid acceleration of many more small-molecule drug discovery projects.

ASSOCIATED CONTENT

Data Availability Statement

The software technologies used for this study are available at <https://www.schrodinger.com/downloads/releases>. For all systems studied, input parameter files, receptor and ligand structures, experimental activities, and output files from the FEP-PB active learning workflow can be found in the Supporting Information.

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.3c00681>.

2D structures of all ligand datasets along with experimental potency data, additional details on the default and top-performing FEP protocols for MCL and p97, additional details on FEP convergence criteria, and the DESMOND relaxation protocol (PDF)

Additional supporting details (ZIP)

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Notes

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